

Vol 8 Chapter 9 — Continuum Elasticity

Keeper (author pass — deep math/physics revision)

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Chapter 9 — Continuum Elasticity

Level 1 — one sentence

Elasticity describes deformable solids: small applied stress σ_{ij} produces proportional strain $\epsilon_{ij} = (1/2)(\partial_i u_j + \partial_j u_i)$ via the elastic-constant tensor C_{ijkl} (Hooke's law generalization), and the resulting wave equations support longitudinal (P) and transverse (S) elastic waves at speeds set by the material's bulk and shear moduli — foundation for seismology, materials engineering, and the Vol 9 Condensed Matter elastic-response chapter.

Level 2 — graduate-physicist precision

9.1 Strain and stress tensors

A displacement field $\vec{u}(\vec{r})$ produces strain tensor:

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

Symmetric 3×3 tensor. Diagonal elements = elongation/compression along axes. Off-diagonal = shear.

Stress tensor σ_{ij} = force per unit area, also symmetric. Linked by Hooke's law:

$$\sigma_{ij} = C_{ijkl} \epsilon_{kl}$$

with C_{ijkl} the elastic-constant tensor (4-index, with various symmetries). For isotropic medium reduces to two independent constants — typically Young's modulus E and Poisson's ratio ν (or bulk K and shear μ).

9.2 Elastic moduli (isotropic case)

- Young's modulus E : stress/strain for uniaxial pull
- Poisson's ratio ν : lateral contraction ratio when stretched
- Shear modulus μ (or G): shear stress/shear strain
- Bulk modulus K : pressure/volumetric strain

Relations: $\mu = E/[2(1 + \nu)]$, $K = E/[3(1 - 2\nu)]$.

Steel: $E \approx 200$ GPa, $\nu \approx 0.3$. Rubber: $E \approx 0.01 - 0.1$ GPa, $\nu \approx 0.5$ (nearly incompressible).

9.3 Equations of motion

Newton's second law applied to a continuum:

$$\rho \partial_t^2 u_i = \partial_j \sigma_{ij} + f_i$$

with ρ density, f_i external body force. Substituting Hooke's law:

$$\rho \partial_t^2 u_i = C_{ijkl} \partial_j \partial_l u_k + f_i$$

For isotropic: $\rho \partial_t^2 \vec{u} = (K + \mu/3) \nabla(\nabla \cdot \vec{u}) + \mu \nabla^2 \vec{u} + \vec{f}$.

9.4 Elastic waves

Decomposing $\vec{u} = \vec{u}_L + \vec{u}_T$ (longitudinal + transverse parts), each satisfies a wave equation:

- P-wave (longitudinal, $\nabla \times \vec{u} = 0$): $v_P = \sqrt{(K + 4\mu/3)/\rho}$
- S-wave (transverse, $\nabla \cdot \vec{u} = 0$): $v_S = \sqrt{\mu/\rho}$

$v_P > v_S$ always. Earthquake seismology: P-waves arrive first, then S-waves; difference in arrival times tells distance to epicenter.

For Earth interior: - P: ~6-13 km/s - S: ~3-7 km/s

S-waves can't propagate through liquids ($\mu = 0$) — used to determine Earth's outer core is liquid.

9.5 Beam bending: Euler-Bernoulli

For a thin beam of length L , Young's modulus E , moment of inertia I , deflection $w(x)$ under load $q(x)$:

$$EI \frac{d^4 w}{dx^4} = q(x)$$

(Euler-Bernoulli equation). Solutions for various boundary conditions give cantilever, simply-supported, etc. Foundation of structural engineering.

9.6 Yield and plasticity

Above the yield stress σ_Y , materials deform plastically — irreversible deformation. Beyond plastic regime, fracture. Engineering uses safety factor 2-3 below σ_Y for designs.

9.7 Substrate connection: phonons and Vol 9

Volume 9 Chapter 7 develops phonons — quantized elastic waves in crystalline solids. The classical elastic-wave equations of this chapter, quantized, give phonon spectra. The substrate's K-type structure at the lattice scale produces the substrate-natural elastic response.

9.8 Worked example: steel beam bending

A steel beam 1 m long, cross-section 1 cm $\times$ 1 cm, supported at both ends, loaded with 100 N at center.

$$I = bh^3/12 = (0.01)(0.01)^3/12 = 8.3 \times 10^{-10} \text{ m}^4 \quad E = 200 \text{ GPa} = 2 \times 10^{11} \text{ Pa} \quad \text{Maximum deflection: } w_{\max} = PL^3/(48EI) = (100)(1)^3/(48 \cdot 2 \times 10^{11} \cdot 8.3 \times 10^{-10}) \approx 0.013 \text{ m} = 1.3 \text{ cm}$$

9.9 K-audit anchors

- Vol 6 Ch 7 (phonons preview)
- Vol 9 Ch 7 (phonons in condensed matter)

Level 3 — 5th-grader accessibility

Solids stretch, compress, and shear when you push them. Strain = how much they deform. Stress = how much you're pushing. Linked by Hooke's law $\sigma = E\epsilon$ (Young's modulus). For isotropic materials, 2 independent constants: $E + \nu$ (Poisson's ratio = how much you contract sideways when stretched). Elastic waves in solids: P-waves (longitudinal, faster) + S-waves (transverse, slower). Seismologists use P-S arrival difference to locate earthquakes. S-waves can't travel through liquids → that's how we know Earth's outer core is liquid. Beam bending: Euler-Bernoulli equation predicts deflection under load — basis of structural engineering. In BST, elastic response is the substrate's atomic K-type configuration response to applied force; phonons (quantized elastic waves) are substrate K-type vibrations.

What comes next

Chapter 10 develops fluid mechanics and Navier-Stokes — including BST's ~99% proved Millennium result.

Where to look this up

- Landau-Lifshitz, *Theory of Elasticity*
- BST: Vol 6 Ch 7; Vol 9 Ch 7